Blind Noise-Template Subtraction with a Fine-Tuned Demucs Denoiser for Indic Speech Noise Removal

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Code, checkpoints and templates: https://github.com/syehsyoh-dokling/NoiseRemoval

***Abstract*—Track 2 asks for Vaani speech recordings with their noise events removed, scored by SI-SDR on synthetic clips plus the change in word error rate of a mandated ASR. Our submission (Codabench 995100) placed 2nd on the Phase-2 leaderboard with a combined score of 21.2379 (SI-SDR 16.770 dB, ΔWER$\times$100 4.468, audit passed). It rests on one observation: the synthetic test clips reuse a library of noise waveforms that, by our evidence, is not the validation library. We discover those waveforms blindly from the residual of a denoiser, subtract the detected copies with a least-squares gain, and run a Demucs denoiser fine-tuned with an SI-SDR loss restricted to event-free target samples on what remains. Template subtraction followed by re-denoising added 8.48 points over the denoiser-only system (12.7544 to 21.2379).**

I. Training Data and Preprocessing

Vaani data. We used the released validation set of Track 2: 695 synthetic clips (noisy mixture, clean reference and noise timestamps) and 1,806 natural noisy clips, all 16 kHz mono. We also used the Gold tier of the Vaani noise-event dataset [1]: 10,772 clips with verified noise-event timestamps. The Phase-2 test set holds 3,178 unlabelled clips. No clean references or timestamps of the test set were accessed. Beyond the test audio itself, which template discovery consumes transductively (Sec. II-C), two pieces of test-side information were used: (a) the leaderboard score of each of our Phase-2 submissions, the model-selection criterion (Sec. III); (b) Codabench's per-clip result page, which lists per-clip SI-SDR for 500 synthetic test clips of its own choosing (it reports the split as 1,099 synthetic, 2,079 natural), used for the per-clip analyses in Sec. III including the ESC-50 decision. Discovery ran on all 3,178 clips without that split.

Noise bank. A synthetic clip is its clean reference plus pasted noise, up to quantisation (outside the pasted spans the residual is about 10^{-4} in amplitude, full scale 1), so mixture minus reference gives 2,213 noise snippets, the only noise source used for training augmentation in the final system.

Augmentation pipeline. Training mixtures are made on the fly. (1) A target recording is drawn; of four random 4 s crops the one with the largest event-free share is kept (redrawn if below 30%). (2) With probability 0.85, 1–3 bank snippets are pasted at random positions. (3) Each snippet's RMS relative to the local target RMS follows a log-normal law centred on the median ratio of the validation mixtures ($\sigma = 0.5$ in natural-log units). (4) Mixture and target are jointly peak-limited to 0.99. Targets are the 1,806 natural validation clips and the Gold clips; the 695 clean references are added as targets from stage 1 on (Sec. II-B).

Event-free masks. Natural and Gold recordings almost always contain real noise events (99.9% of clips; mean coverage 27–30%). Each target gets a mask that is 0 inside annotated events dilated by $\pm$150 ms and 1 elsewhere; clean references are fully 1.

Local validation. Early variants were compared on 200 held-out synthetic validation clips, at the validation SNR and remixed at the test-like SNR. From stage 0 on every validation clip enters training (natural clips as targets, synthetic clips through the bank, clean references as targets from stage 1), so no held-out set remained; moreover the test noise differs from the validation noise (Sec. II-C): the clean-reference variant that won on both held-out sets (mean SI-SDR 23.30 vs. 18.18 dB, test-like SNR) lost on the leaderboard (Table I, row 3). Later comparisons therefore used leaderboard submissions.

Additional datasets. ESC-50 [2] was tested as extra bank noise in one rejected variant (12.0588 vs. 12.3660 without it); it is not used by the final system.

II. Proposed Method

A. Background

SI-SDR [3] projects the estimate onto the reference, so it is gain-invariant; it is $-\mathcal{L}$ in (1) with $m \equiv 1$. The combined score is the mean SI-SDR over the 1,099 synthetic clips plus ΔWER$\times$100, with $\Delta\text{WER} = \text{WER(noisy)} - \text{WER(enhanced)}$ pooled over the words of all 3,178 clips (WER: word edits over reference words). Demucs [4] is a waveform-domain U-Net with an LSTM bottleneck; its dns64 weights, trained on DNS-challenge data [5], are our starting point. Template subtraction rests on the matched filter [6]: for a known waveform T in additive white noise the correlator $\langle x, T\rangle$ is the optimal detector, and normalised cross-correlation [7] makes it gain-invariant. The least-squares gain $g = \langle x, T\rangle / \|T\|^2$ is the projection of the mixture onto T, so $x - gT$ removes an exact copy up to the bias $\langle s, T\rangle / \|T\|^2$ that the speech induces in g; repeating the search on the residual (pass 2) is a batched form of matching pursuit [8]. Unlike REPET [9], which exploits

repetition within one recording, the dictionary here is built across clips from a denoiser's residuals.

B. Denoiser

The denoiser is Demucs in the dns64 configuration [4] (causal U-Net encoder/decoder with an LSTM bottleneck, hidden size 64, 16 kHz), initialised from the public pretrained dns64 weights [10]. The loss is the negative SI-SDR [3] computed only where the event-free mask m of the target s is 1:

$$\mathcal{L} = -10 \log_{10} \frac{\|\alpha\, m \odot s\|^2}{\|\alpha\, m \odot s - m \odot \hat{s}\|^2}, \tag{1}$$

with $\alpha = \langle m \odot \hat{s},\, m \odot s \rangle / \|m \odot s\|^2$ and $\hat{s}$ the denoiser output. Without the mask the model is trained to keep real events, which are present in its "clean" targets; stage 1 (masked) scored 0.55 above stage 0, though other factors changed too (Table I).

Training runs in four stages, all with AdamW (weight decay 10^{-3}), fp16 mixed precision and gradient clipping at norm 5. Stages 0–1 ran on an RTX 4060 Laptop GPU (8 GB) with a one-cycle schedule: (0) 30k steps, batch 8, peak learning rate 10^{-4} (peak at 10% of the run), targets natural + Gold, no masks; (1) 30k steps, batch 8, peak $5 \cdot 10^{-5}$ (peak at 5%), event-free masks, clean references added as targets. Stages 2–3 ran on Kaggle Tesla T4s with a 10-min warm-up and cosine decay over a wall-clock budget: (2) 10.3 h on one T4 (81,482 steps, batch 16, peak $5 \cdot 10^{-5}$; the kernel's second T4 ran the rejected ESC-50 variant); (3) 8.5 h continuation from stage 2 at peak $3 \cdot 10^{-5}$, two seeds (40 and 41) on the two T4s (78,220 and 78,631 steps). At inference the two seeds are averaged at the waveform level. Software: Python 3.13, PyTorch 2.14.0, CUDA 12.6; random seeds 0 (stage 0), 36 (stage 1), 38 (stage 2), 40/41 (stage 3). Compute for these four stages: about 5 laptop GPU-hours and 27.3 T4-hours; the rejected ESC-50 branch cost 10.3 more, and the early runs of Table I rows 2–3 are not counted.

C. Blind noise-template discovery

We measured that pasted noises repeat: of 300 exact validation noise snippets, 71% have a twin with normalised cross-correlation (NCC) ≥ 0.95 in another clip, and within that sample one horn waveform occurs 69 times. The test library appears not to be the validation library: an early submission that subtracted the 2,213 validation snippets wherever they matched the test residuals scored SI-SDR 7.45 dB (combined 5.40), no better than the pretrained dns64 model (SI-SDR 8.20 dB; combined 7.3784, Table I row 1). On the test set the noise is therefore unknown, but the denoiser residual $r = x - \hat{s}$ approximates it: for the strongest 1.5 s window of 300 random stage-2 residuals, 14% have a twin at NCC ≥ 0.9 and 19% at ≥ 0.8. Discovery proceeds on the residuals of the stage-3 two-seed denoiser:

1) *Seeds*: the strongest 1 s window of each clip's residual, strongest first.
2) *Scan*: each seed is correlated with all residuals decimated to 4 kHz (8 CPU workers); hits with NCC ≥ 0.70 are re-aligned at 16 kHz and kept if NCC ≥ 0.75.
3) *Template*: aligned copies, extended by ± 1.5 s around the 1 s seed, are averaged after least-squares gain normalisation; the template support is where copies agree (coherence ≥ 0.5 on 20 ms frames).
4) *Occurrences*: at each hit the gain g is fitted on the mixture, $g = \langle x, T \rangle / \|T\|^2$.
5) *Pass 2*: steps 1–4 are repeated on $x - \sum gT$, using the residual of the denoiser applied to that subtracted mixture, to find noises hidden under the first ones.

The procedure was first checked on the validation set, where references exist, using stage-2 residuals and no denoiser after subtraction: 211 templates, 1,750 occurrences in 659 of the 695 synthetic clips, mean SI-SDR over the 695 clips from -0.33 dB (input) to 13.08 dB; optimistic, because that denoiser had been trained with these very snippets in its bank. On the test set pass 1 found 223 templates with 5,234 occurrences in 1,541 clips and pass 2 added 209 templates and 1,770 occurrences; in total 7,004 occurrences were subtracted in 1,588 clips. One sign of over-detection is visible: the hit density is 3.4 occurrences per hit clip against 2.7 on validation. Since only 1,099 test clips are synthetic, at least 489 natural clips were also modified. The denoiser that follows subtraction limits the damage from spurious hits (Sec. III).

D. Final output and packaging

For clips with occurrences the output is the two-seed denoiser applied to $x - \sum gT$; other clips keep the denoiser-only output. Each waveform is peak-limited to 0.99 and rounded to a multiple of 4 in int16 (quantisation SNR about 68 dB relative to the mean speech level; this reduces, but does not remove, sensitivity to lowest-bit GPU differences) *before* transcription, and the mandated SraVaani-1.0 ASR [11] transcribes exactly the written files, which is what makes the audit mismatch 0%. The organisers' audit re-transcribes the submitted audio and compares it with the submitted transcripts; two early submissions failed it (11–17% mismatch), all rows of Table I passed. The zip (deflate 9) is 924.5 MB, under the 1 GB limit.

E. Pretrained models, cost and compliance

Demucs dns64 from facebookresearch/denoiser v0.1.5 [10] initialises the denoiser; SraVaani-1.0 (ARTPARK-IISc) [11] produces the transcripts, as mandated. On the 3,178 test clips each denoiser seed takes about 6 min per pass on the laptop GPU (two seeds, three passes) and ASR 12 min; discovery is CPU-bound, about 2 h for pass 1 on 8 workers. Audio data used: the Track 2 release, the organisers' Vaani Gold tier [1], ESC-50 [2] (CC BY-NC 3.0) in the rejected variant only, and the DNS-challenge data [5] carried by the dns64 initialisation; nothing else. Demucs code and dns64 weights are CC BY-NC 4.0 and our released checkpoints inherit that licence.

III. Results and Conclusion

Fig. 1 shows how each accepted change moved the Phase-2 score. Denoiser improvements added 2.86 points over the

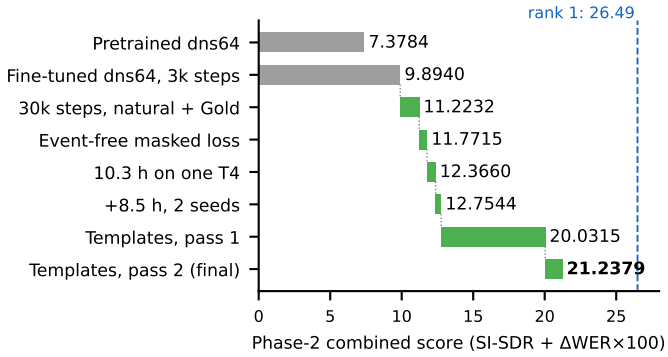


Fig. 1. Phase-2 leaderboard combined score after each accepted change (labels: absolute score). Grey: baselines, each a separate run; green: gain over the previous row. The first green bar compares two separate runs (stage 0 started from the pretrained weights, not from the 3k-step run).

TABLE I
ABLATIONS ON THE PHASE-2 LEADERBOARD (ONE SUBMISSION EACH)

System	Combined
Pretrained dns64, no fine-tuning[a]	7.3784
Fine-tuned 3k steps, natural validation targets	9.8940
Fine-tuned 6k steps, clean refs as targets, test-like SNR[b]	9.0097
Fine-tuned 30k steps, natural + Gold targets (stage 0)[c]	11.2232
+ event-free masked loss, clean refs as targets (stage 1)	11.7715
+ 10.3 h long run (stage 2)	12.3660
+ 8.5 h continuation, 2-seed mean (stage 3)	12.7544
Stage 2 with ESC-50 noise added to the bank (branch)	12.0588
Pass-1 subtraction, hit clips not denoised afterwards[d]	6.1181
Templates pass 1 + denoiser	20.0315
Templates pass 1+2 + denoiser (final)	**21.2379**

[a]No unprocessed-input submission; early near-identity submissions scored SI-SDR 7.4–8.2 dB, so the pretrained model is at input level. [b]From the pretrained weights; test-like SNR ≈7.5 dB, estimated from those submissions. [c]Differs from the 3k-step row in both steps and targets; started from the pretrained weights. [d]Other clips keep the stage-3 denoiser output.

first fine-tuned run (9.8940 to 12.7544; 5.38 over the pretrained model); template subtraction added 8.48 more. Because ΔWER is pooled over words, the split values below do not average to the total. On the leaderboard the final system lifted SI-SDR from 11.751 to 16.770 dB and ΔWER×100 from 1.003 to 4.468, split into synthetic clips (3.01 to 13.19) and natural clips (0.06 to 0.36). Untouched natural clips are identical to the denoiser-only output, so the whole natural gain comes from the ≥489 touched clips: a net gain, though damage to individual clips cannot be excluded. On the 500 listed synthetic clips the final system beats the denoiser-only system on 478, by 4.90 dB on average (the leaderboard mean over all 1,099 synthetic clips rose by 5.02 dB), most on the hardest clips (+6.2 to +6.6 dB where the denoiser alone scored 0–10 dB).

Table I lists the main ablations. Every row is a single submission, so we add per-clip sign tests on the 500 listed clips where we have them (binomial $p < 10^{-6}$ for each): stage 0 beats the 3k-step run on 430 clips, stage 3 beats stage 2 on 459 (+0.18 dB mean), the ESC-50 branch is worse than stage 2 on 344, and the final system beats stage 3 on 478; the stage-1 and stage-2 gains were not tested per clip. Subtraction without denoising the hit clips scores only 6.1181, below the pretrained model: it leaves residue, speech-biased gains, spurious hits, unmatched noises and alignment error, which the following denoiser largely repairs (6.1181 to 20.0315); the two steps are complementary. Stage 3's +0.39 mixes longer training and two-seed averaging (single seeds were not submitted). The clean-reference variant (9.0097) changed three things at once (targets, noise SNR, 6k steps), so its deficit is not attributed to one cause.

Limitations. Template discovery is transductive: it runs on the test audio itself and exploits a property of the organisers' generator (a finite noise library), so it is not a general denoising method; its precision and recall were measured neither on test (no labels) nor on validation, so the over-detection signs in Sec. II-C are the only evidence on false positives. On recordings with unique noise only the denoiser acts (the 12.7544-level system). Model selection used the Phase-2 leaderboard: 20–26 submissions in total (old submission files were deleted to free Codabench storage, so the exact count is not recoverable); the 12 scores quoted here (Table I and the 5.40 run of Sec. II-C) informed design decisions, the rest were packaging or audit re-uploads and early near-identity variants (outputs close to the inputs). Training-run variance was not estimated (the stage-3 seeds were only averaged).

Conclusion. When a test set is generated from a finite noise library, the noise itself can be recovered without labels by correlating denoiser residuals across clips. Exact subtraction of the recovered waveforms, then a learned denoiser, lifted the combined score from 12.7544 to 21.2379, rank 2 (rank 1: 26.4894). The released code, checkpoints and templates regenerated the 24 test clips we checked bit-identically on our laptop GPU (12 with template hits); other GPUs may differ in the lowest fp16 bits.

REFERENCES

[1] ARTPARK, IISc and Google, "Vaani noise event dataset," Hugging Face, https://huggingface.co/datasets/ARTPARK-IISc/Vaani-Noise-Event-Dataset, 2026.
[2] K. J. Piczak, "ESC: Dataset for environmental sound classification," in *Proc. ACM Multimedia*, 2015.
[3] J. Le Roux, S. Wisdom, H. Erdogan and J. R. Hershey, "SDR – half-baked or well done?" in *Proc. ICASSP*, 2019.
[4] A. Défossez, G. Synnaeve and Y. Adi, "Real time speech enhancement in the waveform domain," in *Proc. Interspeech*, 2020.
[5] C. K. A. Reddy *et al.*, "The INTERSPEECH 2020 deep noise suppression challenge: Datasets, subjective testing framework, and challenge results," in *Proc. Interspeech*, 2020.
[6] G. L. Turin, "An introduction to matched filters," *IRE Trans. Inf. Theory*, vol. 6, no. 3, pp. 311–329, 1960.
[7] J. P. Lewis, "Fast normalized cross-correlation," in *Vision Interface*, 1995.
[8] S. G. Mallat and Z. Zhang, "Matching pursuits with time-frequency dictionaries," *IEEE Trans. Signal Process.*, vol. 41, no. 12, pp. 3397–3415, 1993.
[9] Z. Rafii and B. Pardo, "REpeating Pattern Extraction Technique (REPET): A simple method for music/voice separation," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 21, no. 1, pp. 73–84, 2013.
[10] Facebook Research, "denoiser," v0.1.5, https://github.com/facebookresearch/denoiser, 2020.
[11] ARTPARK-IISc, "SraVaani-1.0," https://huggingface.co/ARTPARK-IISc/SraVaani-1.0, 2026.